

Personalised Practice: Coordinate Geometry

Topic: Coordinate Geometry

Original examination-style practice material

Name: _____	Date: _____
Time allowed: 25 minutes	Total marks: 12

Show all necessary working. Give units where appropriate. Answer every question.

1 Points A(-3, 5) and B(7, -1) lie on a straight line.

(a) Find the gradient of AB. [2]

(b) Find the midpoint of AB. [1]

2 A line passes through P(-2, 7) and Q(4, -5). Find its equation in the form $y = mx + c$.

[4]

3 The line L has equation $y = 2x - 1$. A line M is perpendicular to L and passes through (6, 4). Find the equation of M.

[5]

End of Question Paper — check your answers.

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1 (a) $-3/5$ (b) (2, 2)

Gradient = $(-1 - 5) / (7 - (-3)) = -6/10 = -3/5$.

Midpoint = $((-3 + 7)/2, (5 + (-1))/2) = (2, 2)$.

Marks: correct signed differences · correct gradient · correct midpoint [3]

Accept: -0.6 for the gradient

2 $y = -2x + 3$

$m = (-5 - 7)/(4 - (-2)) = -12/6 = -2$.

Using P: $7 = -2(-2) + c$, so $c = 3$.

Marks: gradient method · $m = -2$ · substitution · equation [4]

3 $y = -1/2 x + 7$

Perpendicular gradient = $-1/2$.

$4 = -1/2(6) + c$, so $c = 7$.

Marks: negative reciprocal · uses point · finds intercept · forms equation · correct final form [5]

Accept: $x + 2y = 14$
